

Battery Door Mechanism Design – Posha Spindle

Mechanical Engineering Internship Assignment

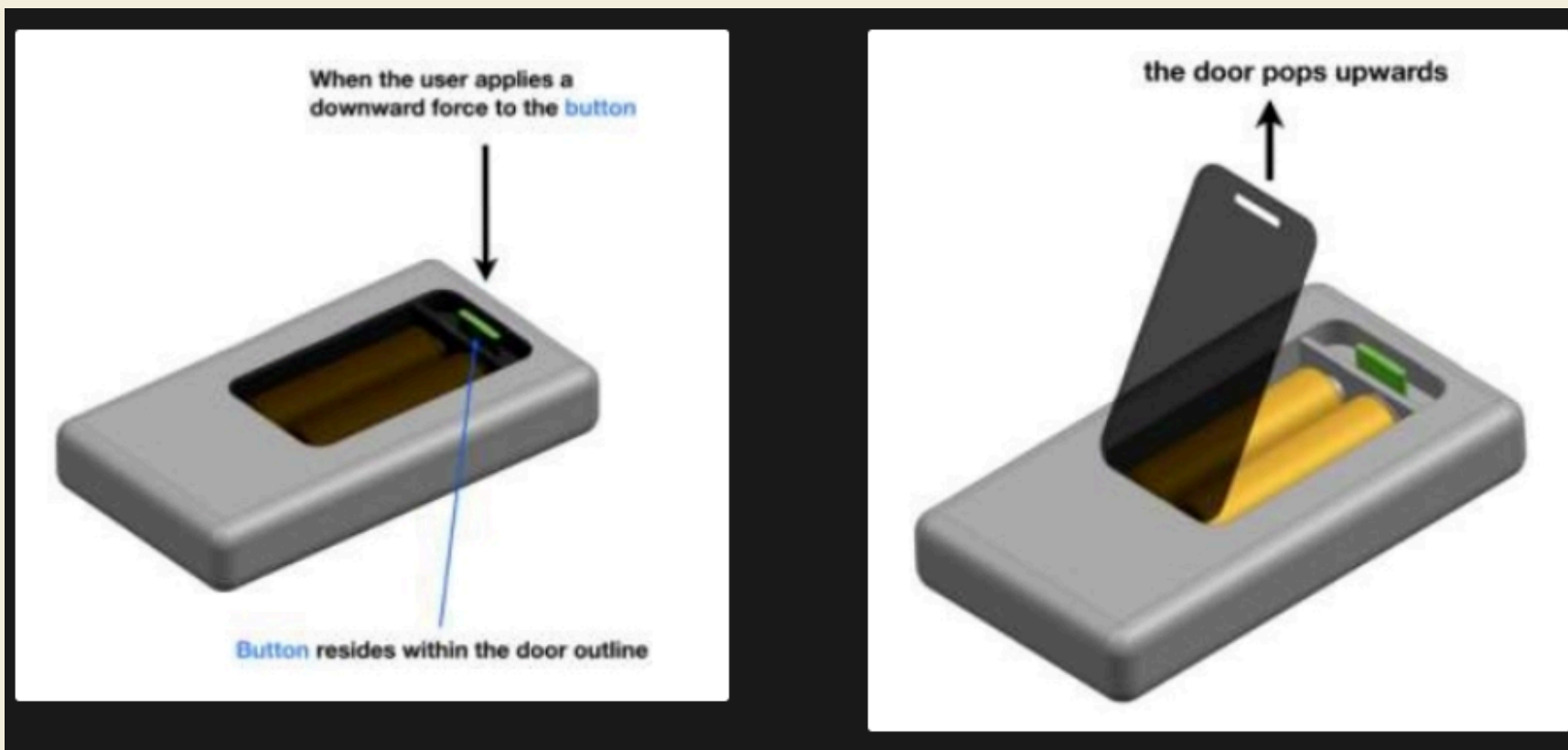
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The Challenge



Objective

Design a battery door mechanism for 2 AA batteries

Key Constraints

Must include a door, button within door outline, and latching system

Input

Downward normal force only

Output

Door automatically pops open in opposite direction

Inspiration & Industry Benchmarking



Digital Cameras

Heavy reliance on auto-locking snap-fits and guided slider systems to minimize space.

TV Remotes

Sliding wedge latches to reduce part count and assembly complexity.

The "Pop-Up" Force

Borrowing from standard industry practices — the design utilizes the constant upward force of standard AA battery contact springs as the automatic ejection mechanism once the latch is cleared.



Concept Exploration & Rejected Ideas

REJECTED CONCEPTS

Concept 1: Multi-Pivot Lever System

1

Rejection Rationale: Excessive pin joints created tolerance stack-up issues and overly complex assembly requirements.

Concept 2: Multi-Wedge Spring System

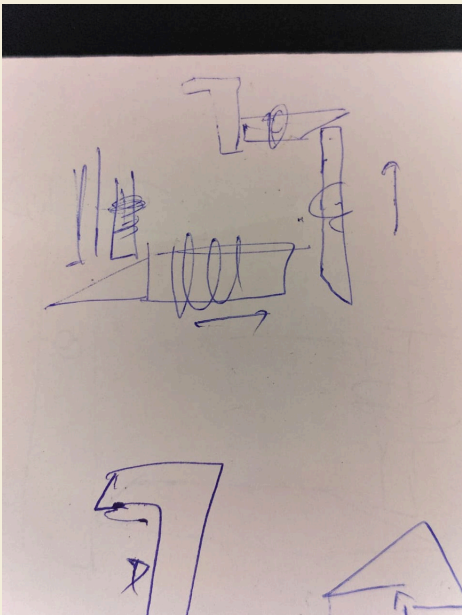
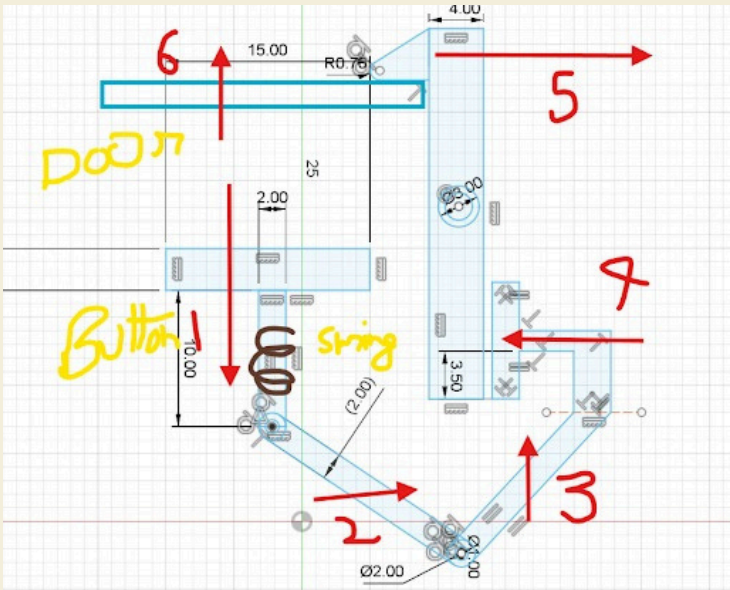
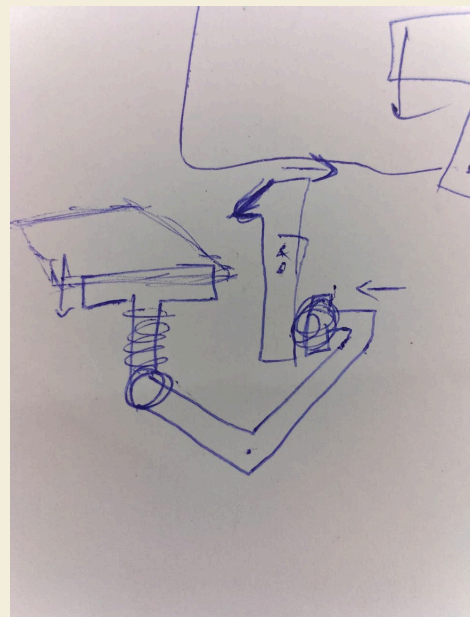
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Rejection Rationale: High spring count caused unstable motion transfer and elevated internal friction.

KEY LEARNING



Reduce degrees of freedom (DOF) and utilize guided linear motion for reliable mechanism performance.



The Compliant Flexure Mechanism

1

Zero-Assembly Design

Compliant mechanism relying entirely on material deformation (flexing)—no moving parts or sliding tracks.

2

Working Principle

Flat plastic beam runs horizontally across the door with integrated latch hooks on both ends. Pushing the center button bends the beam downward into a 'U' shape.

3

Kinematics

Fixed physical beam length forces the two ends to pull inward horizontally as the beam bends, instantly releasing the door catch.

4

Key Advantage

Reduces internal mechanism part count to exactly one, completely eliminating assembly time and cost.



Concept Selection Matrix: Decision Tree for Final Mechanism Selection

Concept	Moving Parts DoF Assembly Risk	Verdict
Multi-Pivot System	5 Parts 5 DoF Complex Assembly High Tolerance Risk	REJECTED
Compliant Flexure	1 Part 1 DoF Zero Assembly High Material Fatigue Risk	REJECTED
Wedge-Driven Sled	3 Parts 2 DoF Simple Drop-in Low Tolerance Risk	SELECTED
Final Rationale	Wedge mechanism ensures long-term durability (no plastic creep) and premium tactile click	WINNER

Wedge mechanism selected for guaranteed durability and superior user experience over flexure design

SOLUTION

Final Mechanism – Wedge Driven System

1

Button Plunger

Captures user input and initiates vertical force transmission

2

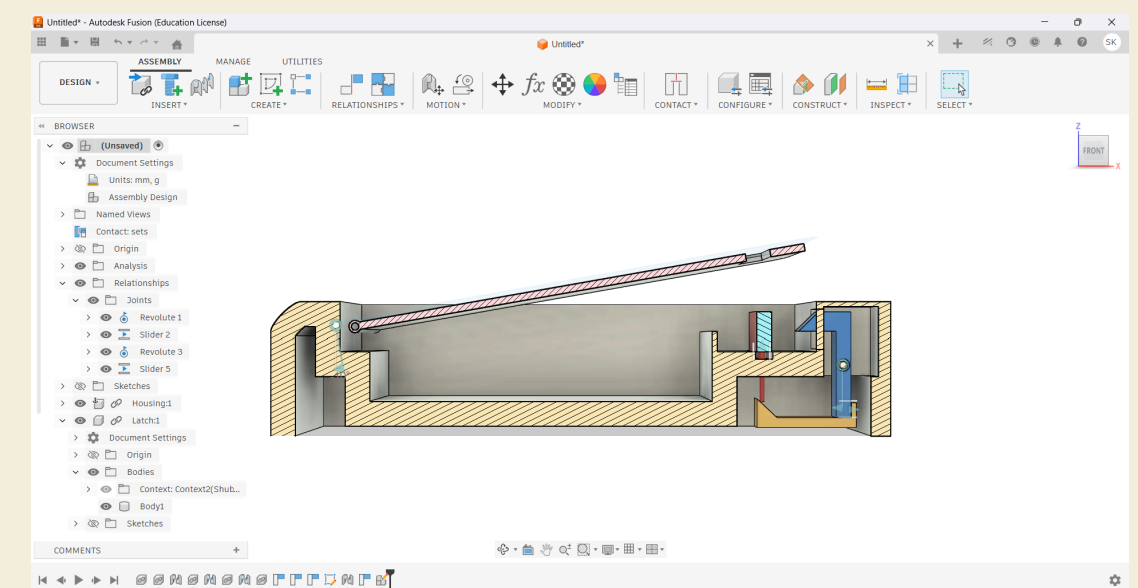
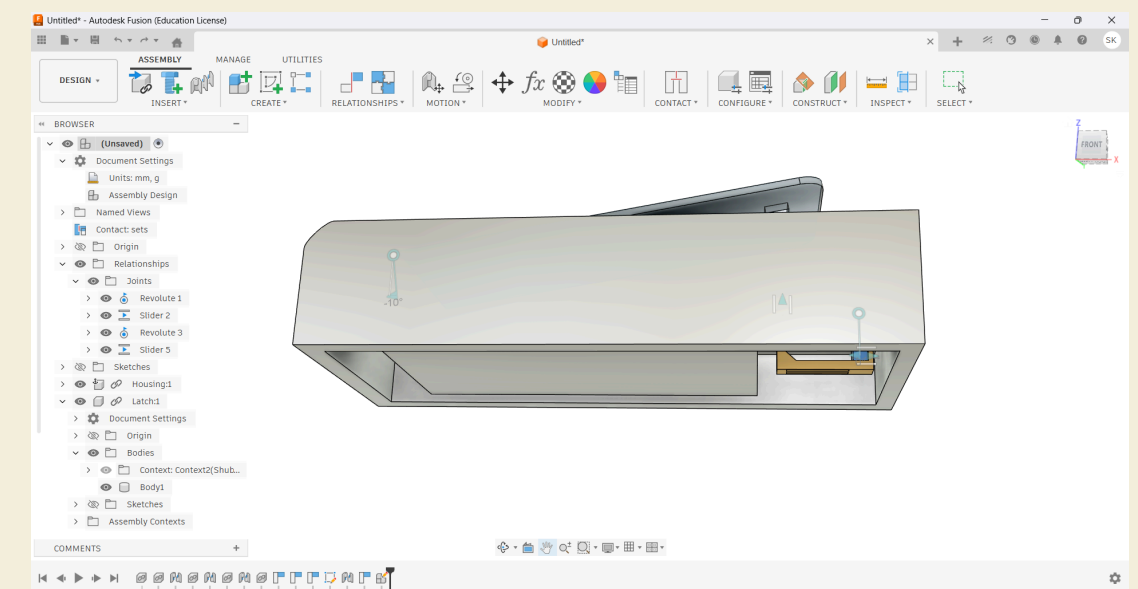
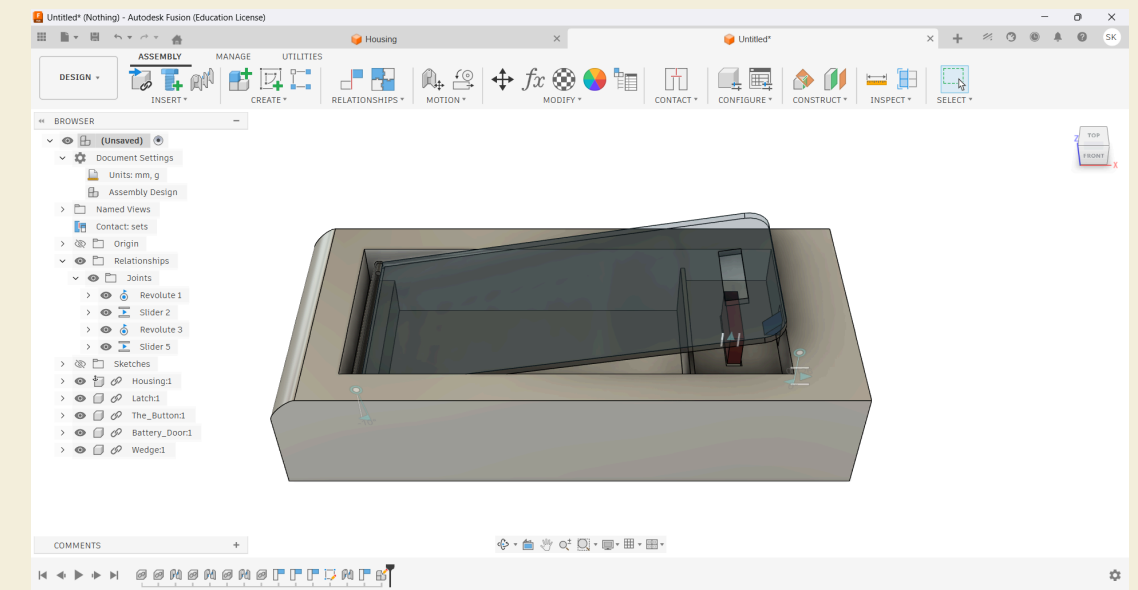
Guided Wedge Sled

Converts vertical force to horizontal motion via guided motion tracks

3

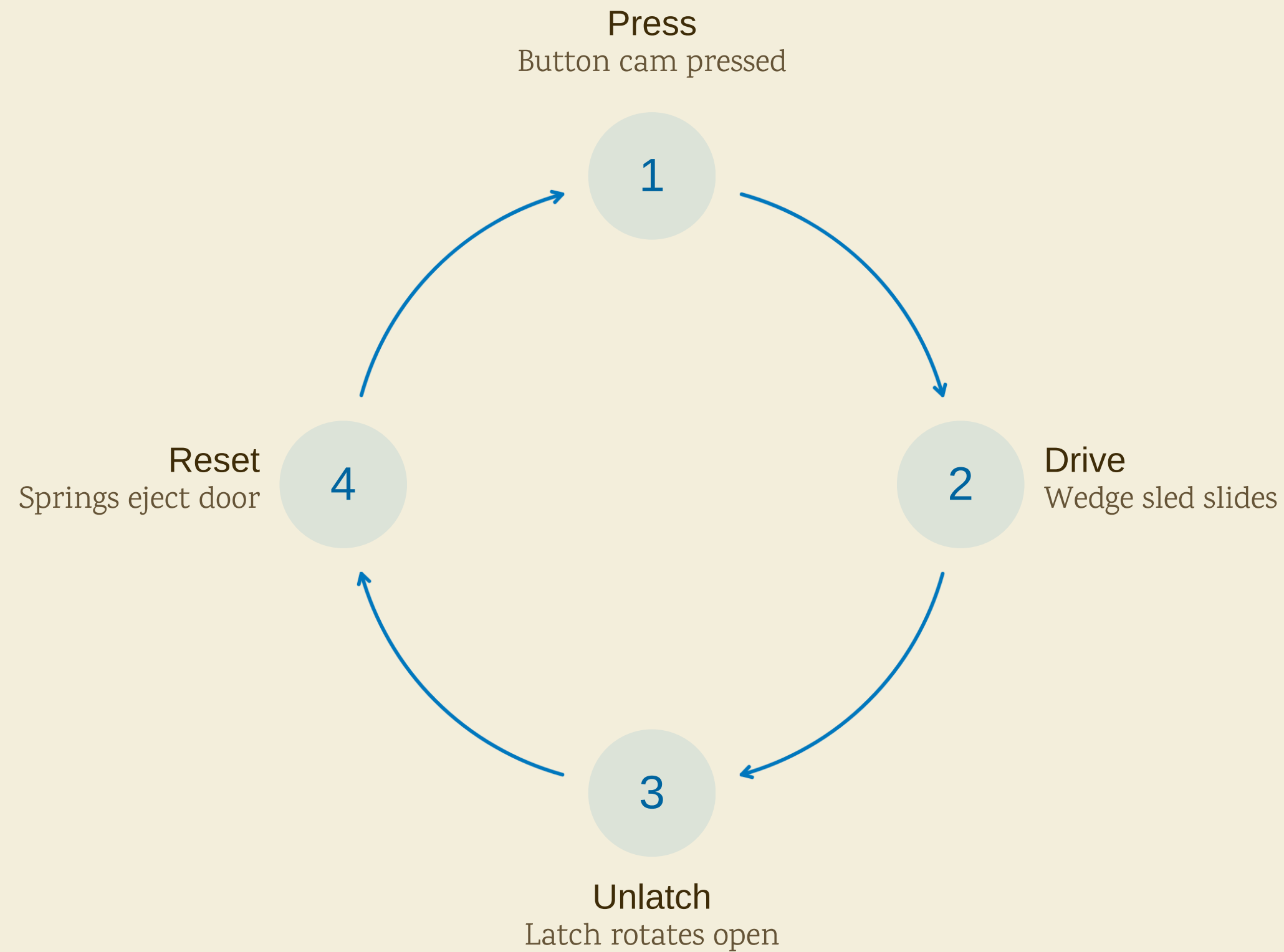
Rotating Latch

Disengages the door catch securely with precise mechanical action



Simplified 3-piece design with guided motion tracks ensures reliability and highly compact internal layout with clear force transmission paths

Working Principle: Activation Cycle



Kinematic & Force Validation

FORCE TRANSLATION

1

45° Wedge Angle: Ensures efficient 1:1 theoretical force transfer between button plunger and slider

2

Calculation: $F_{\text{horizontal}} = F_{\text{downward}} \times \tan(45^\circ)$
(excluding surface friction)

EJECTION CONDITION

1

Required: $F_{\text{BatterySprings}} > m_{\text{door}} \times g + F_{\text{LatchFriction}}$ to overcome door weight and residual latch friction

SPECIFICATIONS

Compact mechanism engineered for precision and reliability

MATERIAL & PROPERTIES

POM/Delrin | 65 MPa | $\mu=0.20$

Polyoxymethylene for sliding components: tensile yield strength ~65 MPa, dynamic coefficient of friction ~0.20 (unlubricated)

FOOTPRINT

15mm × 10mm

Mechanism flush within tight internal boundary, adhering to Spindle door outline constraints

TOLERANCES

± 0.1 mm | 0.15 mm clearance

Injection molding tolerance on sled guideways: ± 0.1 mm; clearance fit prevents binding while minimizing axial play

SPRING RATE

$k = 0.5$ N/mm

Micro-compression coil springs with 0.5 N preload and 4.0 mm max travel for consistent return force

Structural FEA & Validation Strategy

FEA Validation

Structural FEA will be conducted in Ansys to ensure the plastic mechanism survives repeated cycling and validates structural integrity under repeated use.

High-Stress Focus Areas

The pivot point and neck of the rotating latch hook experience the highest bending stress during actuation and are identified as critical areas requiring analysis.

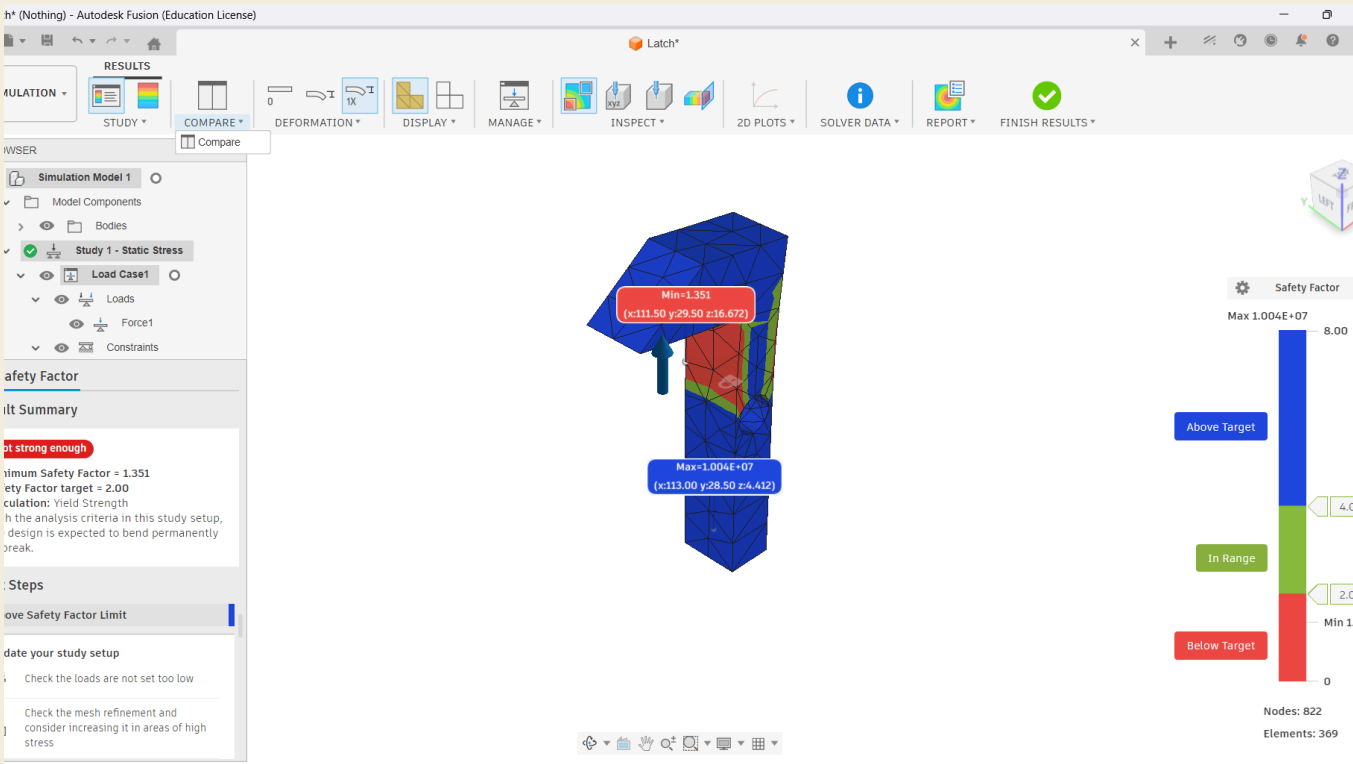
Topology Optimization

Topology optimization will be applied to the wedge sled to remove excess material mass while maintaining the stiffness required to overcome the return springs.

FEA Results & Optimization: Static Structural Analysis Validation

PEAK STRESS	MATERIAL YIELD STRENGTH	FACTOR OF SAFETY	RECOMMENDED FILLET INCREASE
40.95 MPa	~65 MPa	~1.5	+0.5 mm

DFM Optimization: Increase inner fillet radius by 0.5 mm at latch hook to distribute bending force and improve durability. Maximum von-Mises stress (40.95 MPa) concentrated at inner fillet under 15 N horizontal load—primary stress riser identified.



Manufacturing BOM & Materials

Housing & Door: Polycarbonate/ABS blend (PC/ABS) — selected for high impact resistance.

Wedge Sled & Latch: Polyoxymethylene (POM/Delrin) — chosen for exceptionally low coefficient of friction, critical for smooth sliding motion without lubrication.

Return Springs: Standard off-the-shelf micro coil springs (stainless steel).

Assembly Cost: ~\$0.60 per unit (high-volume injection molding).

Conclusion & Future Prototyping

Selected Design

The **Wedge Mechanism** provides controlled motion path, compact integration, and superior force transmission.

Next Step – Prototyping

SLA 3D printing of the 3-piece mechanism to physically validate the tactile feel of the button press.

Next Step – Refinement

Optimize the 45-degree wedge angle based on physical friction testing to improve durability and user feel.

